DATA STRUCTUES HOMEWORK -1

1.Sum of two Matrix.

```
 #include <stdio.h>
int main() {
   int r, c, i, j;
   printf("Enter rows and columns: ");
   scanf("%d%d", &r, &c);
   int A[10][10], B[10][10], C[10][10];
   printf("Enter first matrix:\n");
   for(i=0;i<r;i++){
      for(j=0;j<c;j++){
         scanf("%d",&A[i][j]);
}}
   printf("Enter second matrix:\n");
   for(i=0;i<r;i++){
      for(j=0;j<c;j++){
        scanf("%d",&B[i][j]);
}}
   printf("Sum Matrix:\n");
   for(i=0;i<r;i++) {
      for(j=0;j<c;j++) {
         sum[i][j]=A[i][j]+B[i][j];
         printf("%d ",C[i][j]);
      }
      printf("\n");
   }
   return 0;
```

```
}
```

OUTPUT:

```
Enter rows and columns: 2
2
Enter first matrix:
1
2
3
4
Enter second matrix:
1
2
3
3
Sum Matrix:
2 4
6 7
```

---

2. Reverse an Array

```
#include <stdio.h>
int main() {
    int n,i;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    int a[100],rev[100];
    printf("Enter elements:\n");
    for(i=0;i<n;i++)
```

```
        scanf("%d",&a[i]);

    printf("Reverse Order:\n");
    for(i=n-1;i>=0;i--){
        rev[i]=a[i];
        printf("%d ",rev[i]);
}
    return 0;
}
```

OUTPUT:

Enter number of elements: 3

Enter elements:

1

2

3

Reverse Order:

3 2 1

---

3. Count Duplicate Elements

```
 #include <stdio.h>
int main() {
    int n,i,j,count=0;
    printf("\n Enter number of elements :");
    scanf("%d",&n);
    int a[100];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++) {
```

```
        scanf("%d",&a[i]);
    }

    for(i=0;i<n;i++) {

        for(j=i+1;j<n;j++) {
            if(a[i]==a[j]) {
                count++;
            }
        }
    }

    printf("Duplicate Elements = %d",count);
    return 0;
}
```

OUTPUT:

```
Enter number of elements :5
Enter the array Elements :
1
1
2
2
3
Duplicate Elements = 2
```

---

4. Print Unique Elements

```
 #include <stdio.h>
```

```
int main() {
    int n,i,j,c;
    printf("\n Enter number of elements :");
    scanf("%d",&n);
    int a[100];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Unique Elements:\n");

    for(i=0;i<n;i++) {
        flag=0;
        for(j=0;j<n;j++) {
                if (a[i]==a[j]) {
                    c++;
                } }}
        if(c==1)
            printf("%d ",a[i]);
    }
    return 0;
}
```

OUTPUT:

```
Enter number of elements :5
Enter the array Elements :
1
2
1
```

3

4

Unique Elements:

2 3 4

---

5. Insert into Sorted Array

```
 #include <stdio.h>
int main() {
    int n,i,element;
    printf("\n Enter number of elements:");
    scanf("%d",&n);
    int a[100];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);}
    printf("Enter element to be inserted: ");
    scanf("%d",&element);
     for(i=n;i>=0&&a[i]>element;i--){
        a[i]=a[i-1];}
    a[i+1]=element;
    n++;
  printf("After insertion:");
    for(i=0;i<n;i++){
        printf("%d ",a[i]);}
    return 0;
}
```

OUTPUT:

```
 Enter number of elements:5
Enter the array Elements :
10
20
30
50
60
Enter element to be inserted: 40
After insertion:10 20 30 40 50 60
```

6.Insert into Unsorted Array

```
 #include <stdio.h>
int main() {
    int n,i,pos,element;
    printf("\n Enter number of elements :");
    scanf("%d",&n);

    int a[100];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);}

    printf("Enter position to be inserted: ");
    scanf("%d",&pos);

    printf("Enter the element to be inserted: ");
    scanf("%d",&element);
```

```
    for(i=n-1;i>=pos-1;i--){
        a[i+1]=a[i];}

    a[pos-1]=element;
    n++;

    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

OUTPUT:

```
Enter number of elements :5
Enter the array Elements :
1
2
3
4
5
Enter position to be inserted: 2
Enter the element to be inserted: 6
1 6 2 3 4 5
```

7.Matrix Multiplication

```
#include <stdio.h>
int main()
{
```

```
int r1, c1, r2, c2, i, j, k;
printf("Enter rows and columns of first matrix: ");
scanf("%d%d", &r1, &c1);

printf("Enter rows and columns of second matrix: ");
scanf("%d%d", &r2, &c2);
if (c1 != r2)
{
   printf("Matrix multiplication cannot be done.");
   return 0;}
int a[100][100], b[100][100], c[100][100];
printf("Enter first matrix elements:\n");
for (i = 0; i < r1; i++){
   for (j = 0; j < c1; j++){
      scanf("%d", &a[i][j]);
   }
}
printf("Enter second matrix elements:\n");
for (i = 0; i < r2; i++){
   for (j = 0; j < c2; j++)  {
      scanf("%d", &b[i][j]);
   }
}
for (i = 0; i < r1; i++){
   for (j = 0; j < c2; j++) {
      c[i][j] = 0;
      for (k = 0; k < c1; k++)  {
         c[i][j] += a[i][k] * b[k][j];
      }
```

```
        }
    }
    printf("Result:\n");
    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++){
            printf("%d ", c[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

Enter rows and columns of first matrix: 2

2

Enter rows and columns of second matrix: 2

2

Enter first matrix elements:

2

2

1

3

Enter second matrix elements:

1

3

1

4

Result:

4 14

4 15

8.Matrix Transpose

```
#include <stdio.h>
int main()
{
   int r, c, i, j;
   printf("Enter number of rows and columns: ");
   scanf("%d%d", &r, &c);

   int a[100][100],b[100][100];
   printf("Enter matrix elements:\n");
   for (i = 0; i < r; i++) {
      for (j = 0; j < c; j++) {
         scanf("%d", &a[i][j]);
         b[j][i]=a[i][j];
      }
   }
   printf("Transpose Matrix:\n");
   for (i = 0; i < c; i++) {
      for (j = 0; j < r; j++){
         printf("%d ", b[i][j]);
      }
      printf("\n");
   }
   return 0;
}
```

OUTPUT:

Enter number of rows and columns: 2

2

Enter matrix elements:

7

9

8

1

Transpose Matrix:

7 8

9 1

---

9. Right Diagonal Sum

```
#include <stdio.h>
int main()
{
   int n, i, j, sum = 0;
   printf("Enter the order of matrix: ");
   scanf("%d", &n);
   int a[10][10];
   printf("Enter matrix elements:\n");
   for (i = 0; i < n; i++){
      for (j = 0; j < n; j++) {
         scanf("%d", &a[i][j]);
      }
   }
   for (i = 0; i < n; i++){
```

```
        sum += a[i][i];
    }
    printf("Right diagonal sum = %d", sum);
    return 0;
}
```

OUTPUT:

Enter the order of matrix: 2

Enter matrix elements:

3

1

4

8

Right diagonal sum = 11

---

10. Check Identity Matrix

```
#include <stdio.h>
int main()
{
    int n, i, j, flag = 1;
    printf("Enter the order of matrix: ");
    scanf("%d", &n);
    int a[10][10];
    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++){
        for (j = 0; j < n; j++){
            scanf("%d", &a[i][j]);
        }
```

```
    }
    for (i = 0; i < n; i++){
        for (j = 0; j < n; j++){
            if (i == j && a[i][j] != 1){
                flag = 0;
            }
            if (a[i][j] != 0){
                flag = 0;
            }
        }
    }
    if (flag == 1)
        printf("Identity Matrix.");
    else
        printf("Identity Matrix.");
    return 0;
}
```

OUTPUT:

Enter the order of matrix: 2

Enter matrix elements:

1

0

0

1

Identity Matrix.

11.Count Vowels, Consonants, Digits and White Spaces

```
#include <stdio.h>
int main()
{
   char str[100];
   int i, vowels = 0, consonants = 0, digits = 0, spaces = 0;
   printf("Enter a string: ");
  scanf("%[^\n]", str);

   for (i = 0; str[i] != '\0'; i++){
      if ((str[i] >= 'A' && str[i] <= 'Z') || (str[i] >= 'a' && str[i] <= 'z')){
         char ch = str[i];
         if (ch >= 'A' && ch <= 'Z')
            ch = ch + 32;
         if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
            vowels++;
         else
            consonants++;
      }
      else if (str[i] >= '0' && str[i] <= '9'){
         digits++;
      }
      else if (str[i] == ' '){
         spaces++;
      }
   }
   printf("Vowels = %d\n", vowels);
   printf("Consonants = %d\n", consonants);
```

```
    printf("Digits = %d\n", digits);
    printf("White Spaces = %d\n", spaces);
    return 0;
}
```

OUTPUT:

Enter a string: jothimani 1810

Vowels = 4

Consonants = 5

Digits = 4

White Spaces = 1

12.Frequency of a Character

```
#include <stdio.h>
int main()
{
    char str[100], ch;
    int i, count = 0;
    printf("Enter a string: ");
    scanf("%[^\n]", str);

    printf("Enter the character: ");
    scanf(" %c", &ch);
    for(i = 0; str[i] != '\0'; i++){
        if(str[i] == ch){
            count++;
        }
    }
```

```
    printf("Frequency of  %c = %d", ch, count);
    return 0;
}
```

OUTPUT:

Enter a string: aabc

Enter the character: a

Frequency of a = 2

---

13. Count Words in a String

```
#include <stdio.h>
int main()
{
    char str[100];
    int i, count = 1;
    printf("Enter a string: ");
    scanf("%[^\n]", str);

    for(i = 0; str[i] != '\0'; i++){
        if(str[i] == ' '){
            count++;
        }
    }
    printf("Number of words = %d", count);
    return 0;
}
```

OUTPUT:

```
Enter a string: jothi mani
Number of words = 2
```

---

14. Sort String Array

```
#include <stdio.h>
#include <string.h>
int main()
{
   int n, i, j;
   char str[20][50], temp[50];
   printf("Enter the number of strings: ");
   scanf("%d", &n);

   printf("Enter the strings:\n");
   for(i = 0; i < n; i++){
      scanf(" %[^\n]", str[i]);
   }

   for(i = 0; i < n - 1; i++){
      for(j = i + 1; j < n; j++){
         if(strcmp(str[i], str[j]) > 0){
            strcpy(temp, str[i]);
            strcpy(str[i], str[j]);
            strcpy(str[j], temp);
         }
      }
   }
```

```
    printf("Strings in Sorted Order:\n");
    for(i = 0; i < n; i++){
        printf("%s\n", str[i]);
    }
    return 0;
}
```

OUTPUT:

```
Enter the number of strings: 3
Enter the strings:
sri
akil
mathesh
Strings in Sorted Order:
akil
mathesh
sri
```

---

15. Toggle Case of a Sentence

```
#include <stdio.h>
int main()
{
    char str[100];
    int i;
    printf("Enter a sentence: ");
    scanf("%[^\n]", str);

    for(i = 0; str[i] != '\0'; i++){
```

```
        if(str[i] >= 'A' && str[i] <= 'Z'){
            str[i] = str[i] + 32;
        }
        else if(str[i] >= 'a' && str[i] <= 'z'){
            str[i] = str[i] - 32;
        }
    }

    printf("Sentence after toggling:\n%s", str);
    return 0;
}
```

OUTPUT:

Enter a sentence: Jothimani

Sentence after toggling:

jOTHIMANI
